

Cover vs. stego across operating points — imperceptibility $\leftrightarrow$ robustness trade-off

Cover QR
(no hidden data)



Imperceptible
(clean-trained)
PSNR 54.9 dB · SSIM 1.000 · public ✓



residual $\times 127$ (max ± 0.01)

Robust
(distortion-trained)
PSNR 22.5 dB · SSIM 0.988 · public ✓



residual $\times 5$ (max ± 0.50)

Hybrid robust
(structural mask)
PSNR 20.4 dB · SSIM 0.980 · public ✓



residual $\times 5$ (max ± 0.50)

top: stego QR
(public still scans)

bottom: $|\text{stego} - \text{cover}|$
normalized to show
where bits are embedded

